Sequential Logic – Part 5
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Review

- We have looked at flip-flops and registers (built using flip-flops).
- In this part, we will look at another application of flip-flops, namely to perform counting.
- We can also look at counting as a special type of addition operation.

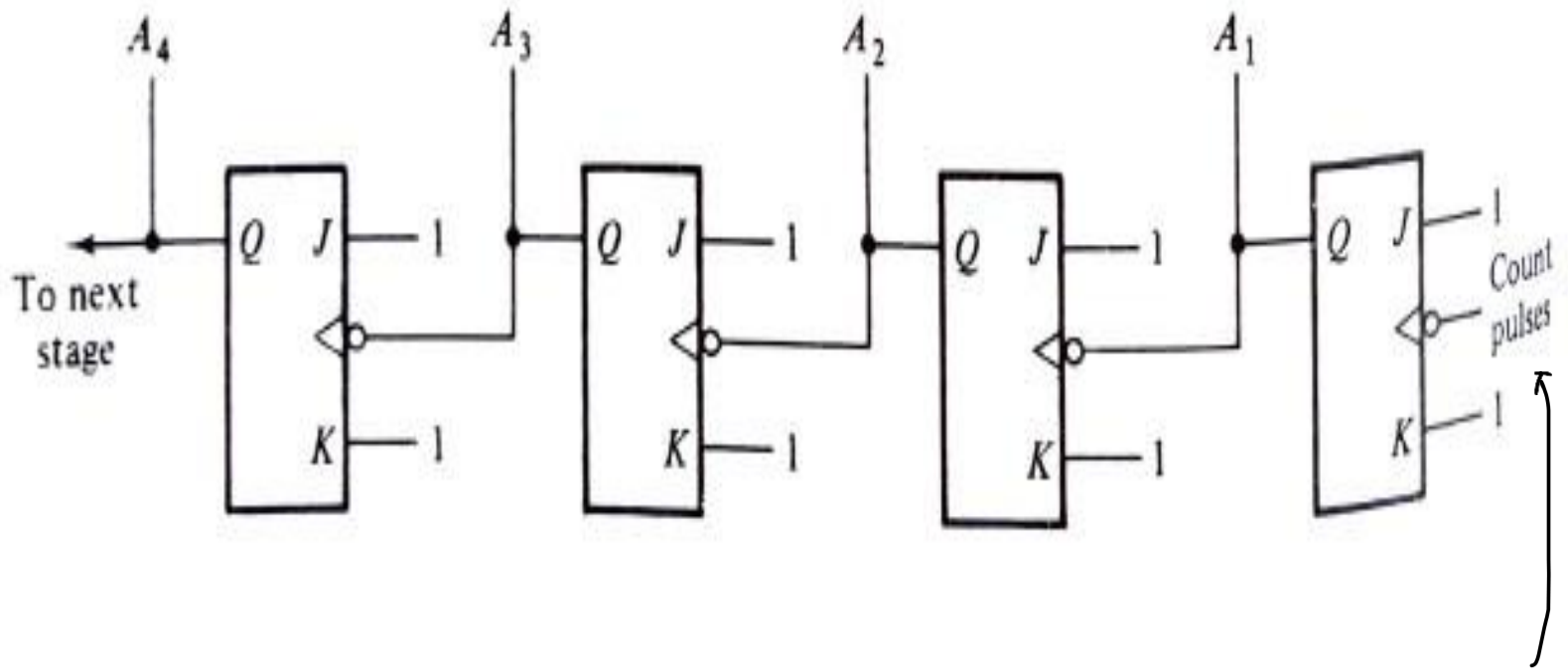
Counters

- These are of different kinds.
- First, we will discuss something called as a ripple counter (or asynchronous counter).
- Then we will discuss a few other types of counters.

What are counters made up of ?

- The simplest counter circuits can be realized using T or J-K flip-flops.
- The toggle feature in T flip-flops is very much suited to implement counting operation. Recall that when J and K are tied together, a J-K flip-flop becomes a T flip-flop.
- In this part, we will only depict the structure of some counters (design will be taken up in the next topic, namely state machines).

Ripple Counter

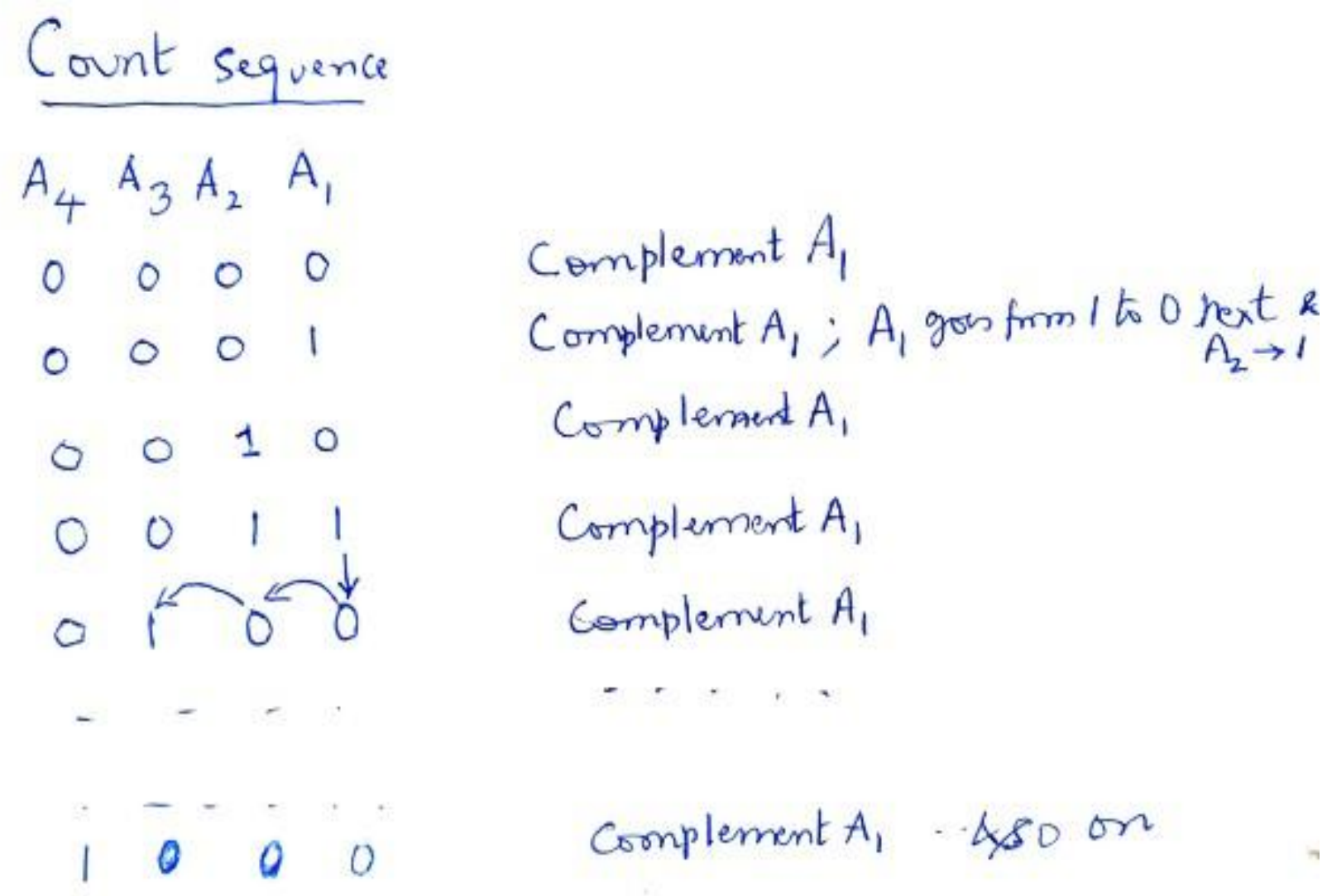


Count pulses

Explanation

- The operation of the ripple counter is as follows. Note that only count pulses are given and there is no common clock (or input pulse) for the four J-K flip-flops here.
- Instead, the output A_1 of the first J-K (on the right) drives the “clock input” of the second J-K. The output of the second drives the “clock input” of the third and so on.
- As a result, the flip-flops change one at a time and the signal propagates through the counter in a ripple fashion. It is asynchronous since triggering of all flip-flops (except the first) is by the transition happening in other flip-flops.

Count sequence



More points on the ripple counter

- The count sequence shows that the lowest order bit, namely A_1 must be complemented with each count pulse.
- Every time A_1 goes from 1 to 0, it complements A_2 .
- Similarly, every time A_2 goes from 1 to 0, it complements A_3 and so on.
- The counter will go on until we reach 1111 and then will repeat from 0.

How to down count ?

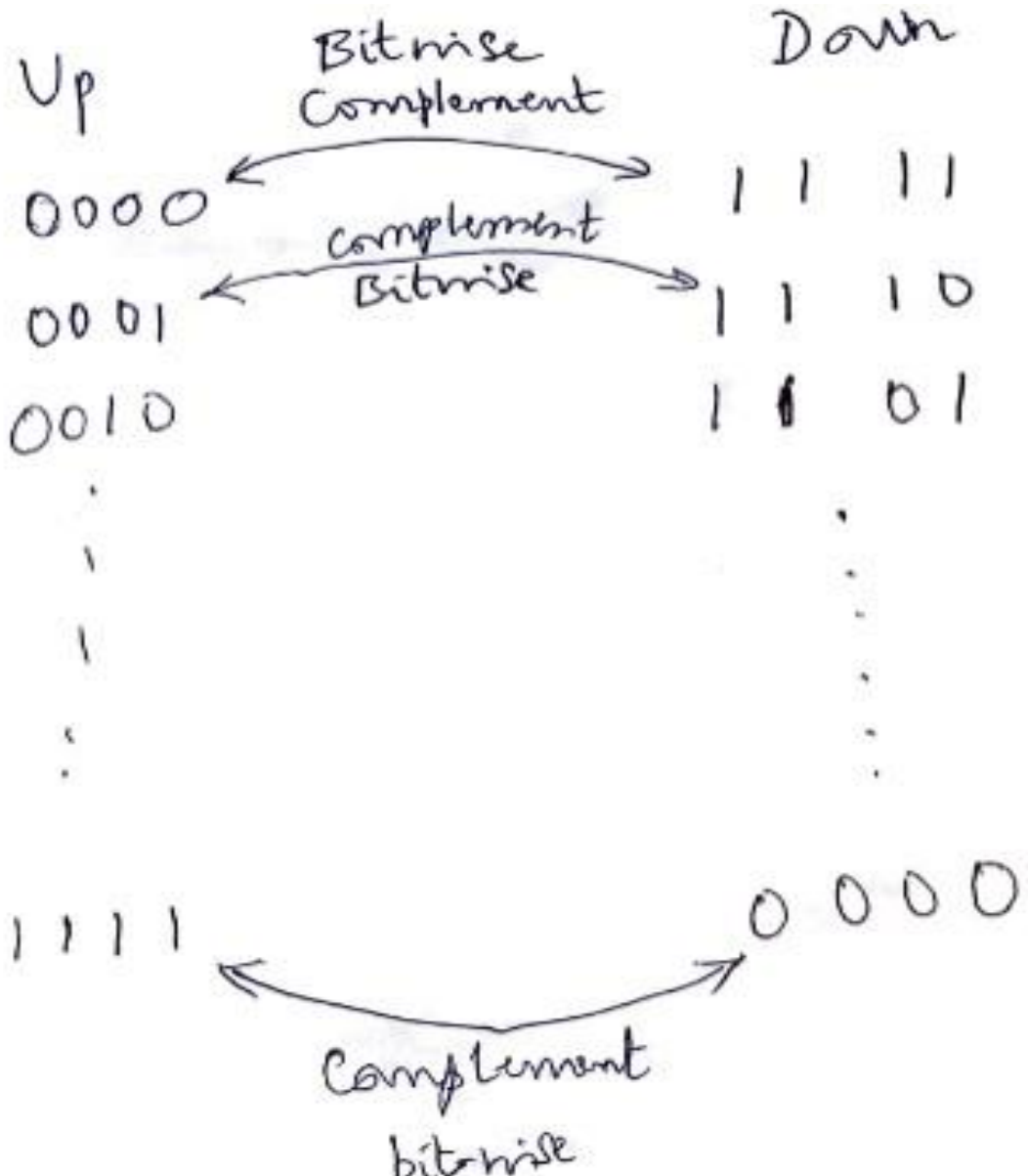
- Question for students.

How to count from 15 to 0 ?

- The same circuit can be used but the outputs need to be taken from the complement terminals, namely Q’ of the flip-flops.

- Some reasoning next.

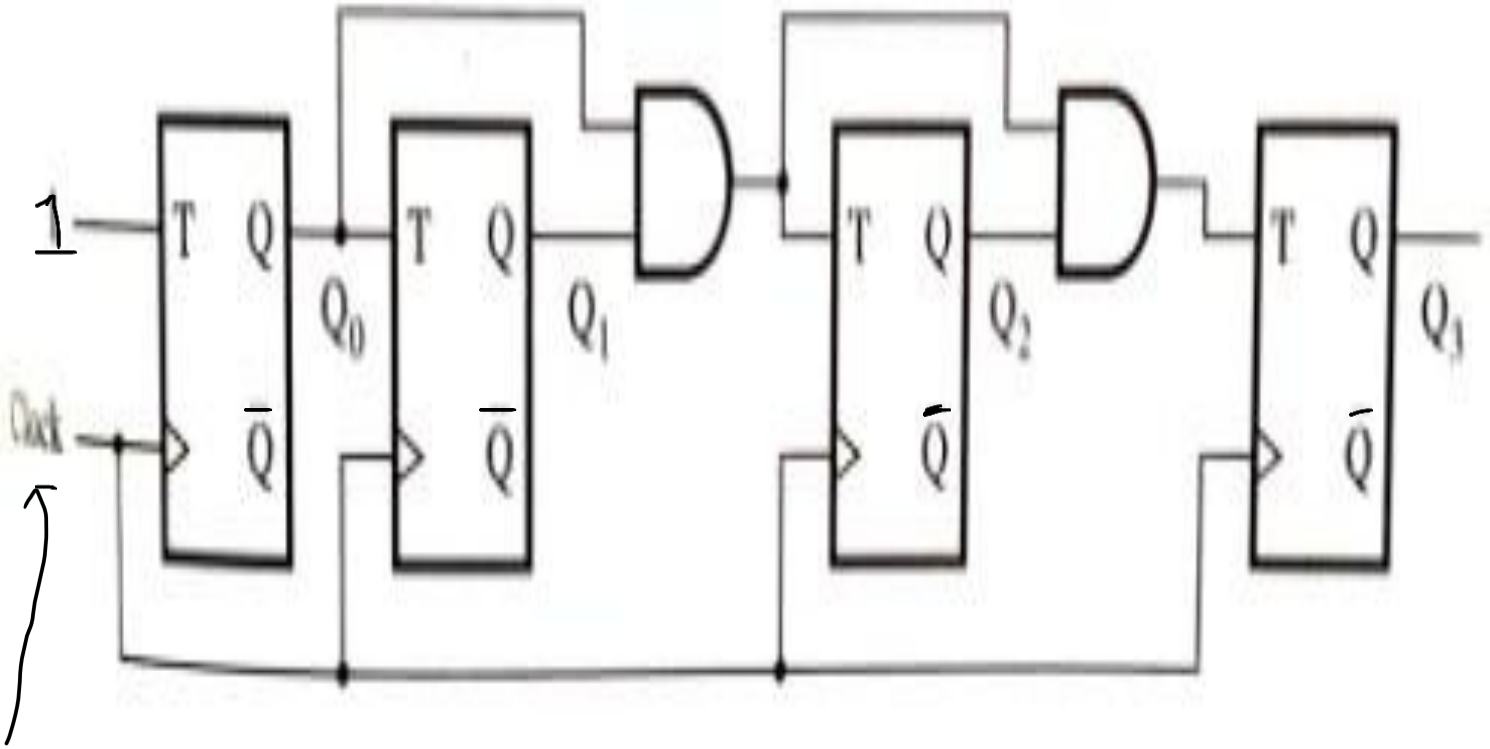
Counting Up versus Counting Down …



Synchronous counters

- The asynchronous counter discussed so far is simple but not very fast.
- Notice that the state change of one flip-flop triggers another flip-flop. This leads to some delay.
- We can instead speed up by clocking all flip-flops at the same time. That is, the operation of the flip-flops is synchronized by a common clock pulse so that when several flip-flops must change state, the state changes occur simultaneously.
- A 4-bit synchronous up-counter is depicted next.

Four-bit synchronous up-counter

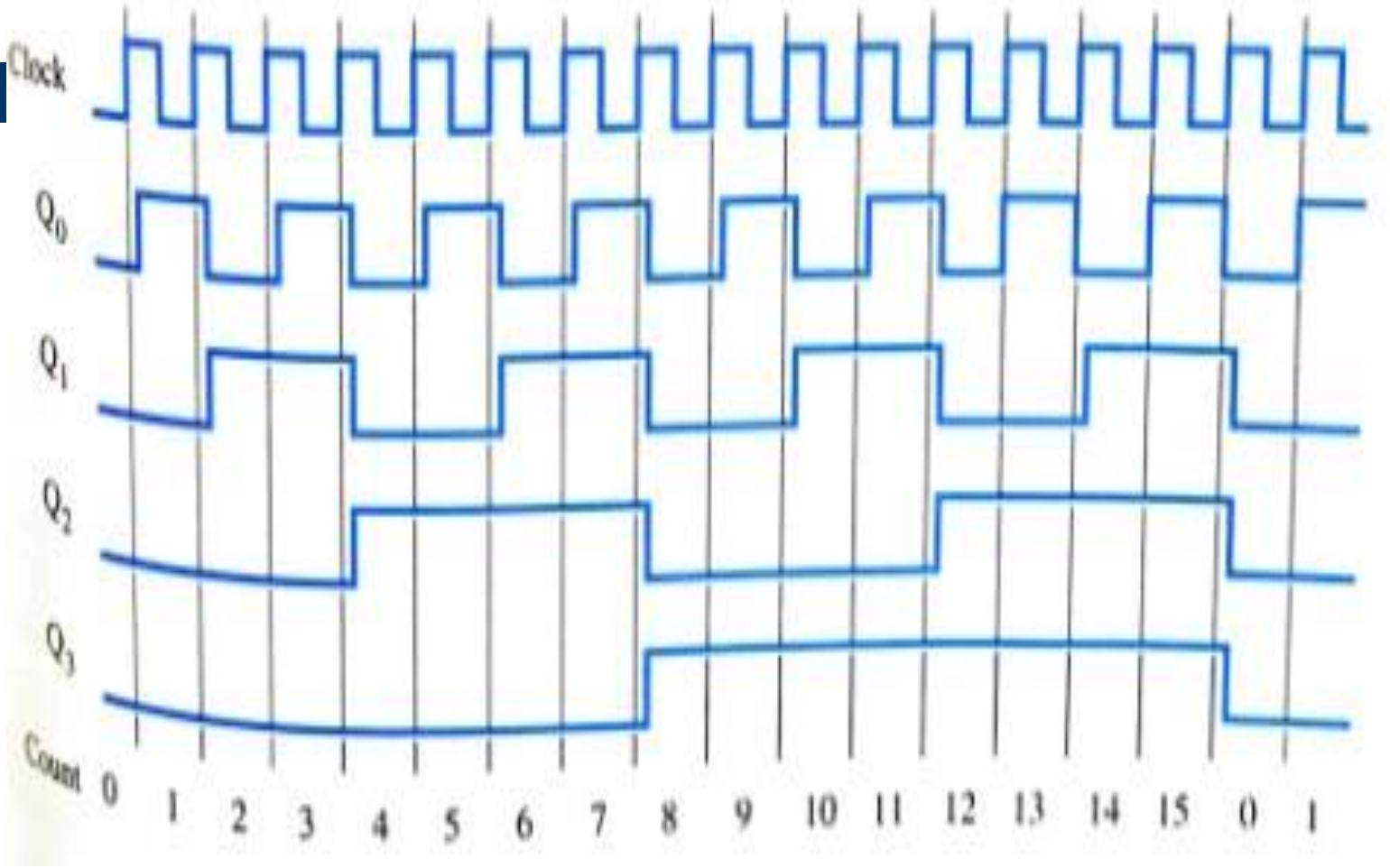


Explanation

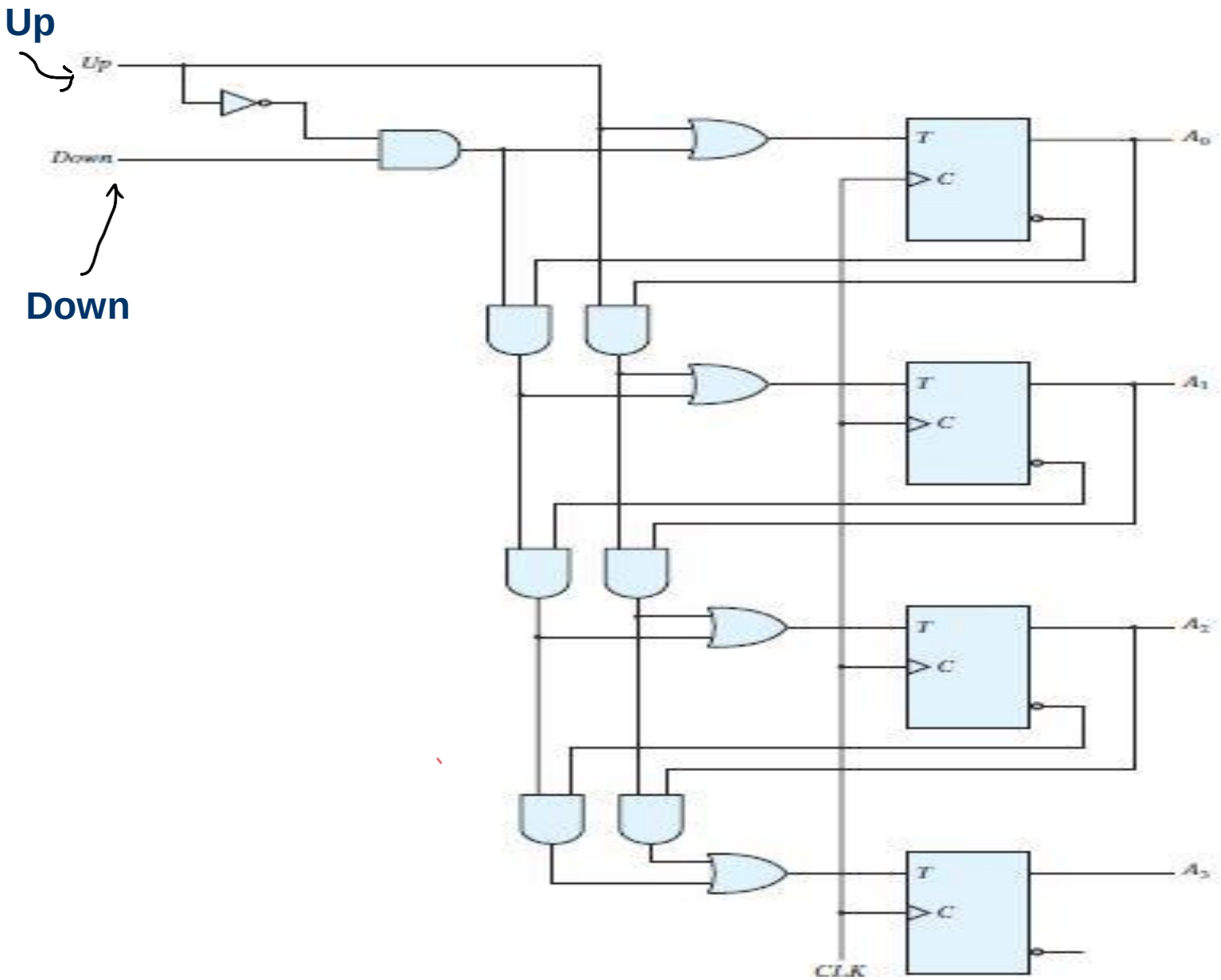
- We have used T flip-flops and clearly $T_0 = 1$,

  $T_1 = Q_0$, $T_2 = Q_0 Q_1$, $T_3 = Q_0 Q_1 Q_2$
- Note that we have a factored arrangement to give the T input to the subsequent stages instead of AND gates with increased size for each stage (which can lead to fan-in problems).
- The circuit will behave as a modulo-16 up-counter.
- The timing diagram is shown next.

Timing diagram



Up-Down Synchronous counter.



Explanation

- The up-down counter combines in one circuit the possibility of counting either up or down.
- If the Up control input is 1 (and the Down input is 0), then the circuit counts up since the T inputs receive their signals from the values of the previous normal (uncomplemented) outputs of the flip-flops.
- However, if the Down input is 1 and the Up input is 0, the circuit counts down since the complemented outputs of the previous flip-flops are applied to the T inputs.

Question for students

- What happens for the other two cases (Up and Down being 0 or both being 1) ?

Question for students

- What happens if both the inputs are 0 ?
- What happens if both are 1 ?

Verify

- If both the inputs are 0, the circuit does not change state and remains in the same count.
- If both the inputs are 1, the circuit counts up.

Remarks

- So far, we have looked at counters composed of T flip-flops or JK flip-flops (with J and K set to 1).
- In particular, the toggle behaviour of T flip-flop helps to realize counters easily.
- A formal procedure to design a counter will be given in the next topic on state machines. Design of sequential circuits for counting-like applications using merely D flip-flops (instead of T flip-flops or JK flip-flops) is somewhat more involved.

Textbook and References

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.
- C. Roth and L. Kinney, Fundamentals of logic design, Cengage Learning, 2014.